

ArgorixLang: Compiled Governance, Sovereign Agent Metadata, and Offline-Verifiable Runtime Evidence

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Abstract

Agent runtimes mix identity metadata, provider permissions, policy decisions, and audit evidence, but these layers are often distributed across prompts, configuration, application code, and logs. This paper presents **ArgorixLang** as a systems artifact for *compiled governance*: a language/runtime boundary in which agent declarations, passports, provider contracts, executable-provider permissions, runtime controls, and evidence artifacts are compiled and checked together. The paper deliberately separates declaration from verification, digest integrity from approval, local passport metadata from authenticated identity, provider contracts from executable adapters, and EvidenceBundles from external attestation. We evaluate the current artifact through a reproducible snapshot of 33 generated request directories. Twenty-seven directories contain complete bytecode, trace, security-report, and Evidence-Bundle artifacts; six contain source only. The 27 complete bundles pass offline internal-consistency verification for bytecode, trace, security report, and a ledger digest recomputed from trace events. The snapshot is structurally repeated, so its 7,155 event records and 1,188 detailed policy findings are evidence of deterministic artifact production rather than workload diversity. A negative finding is central: all complete v0.1 policy reports require review and encode their detailed findings as “unknown policy rule,” showing that digest success does not imply policy approval. We therefore specify the required v0.2 extension: a typed policy decision lattice with pass, deny, review, unknown-rule, and malformed-policy outcomes, plus source-digest binding. We also define controlled cases, ablations, and baselines that must be generated before broader claims are made. The current paper claims a compiled governance and offline evidence boundary; it does not claim security certification, real identity verification, live ANS/DNS, cryptographic attestation, production sandboxing, or large-scale empirical coverage.

Keywords: agent programming languages; compiled governance; sovereign identity; runtime evidence; fail-closed execution; reproducibility

1 Introduction

Modern agent systems compose identities, tools, model providers, delegated permissions, jurisdictional claims, policy decisions, and evidence artifacts. In many deployments those layers live in different places: prompts describe intent, configuration names providers, application code enforces some checks, logs record partial outcomes, and documents describe governance after the fact. This separation is operationally convenient, but it makes a systems question hard to answer: what can be represented, restricted, and evidenced together before an agent operation reaches an executable provider?

ArgorixLang answers this question as a compiled governance runtime rather than as a certification framework. It compiles agent declarations, typed messages, local Agent Passport metadata, provider contracts, policies, runtime profiles, and evidence requirements into bytecode that is executed by a fail-closed VM [3]. The VM separates a declarative provider contract from an executable provider adapter.

A contract may describe an external provider; it does not become callable unless an executable adapter is registered. In the artifact studied here, `simulated` is the only executable provider.

The central thesis is claim-bounded: `ArgorixLang` demonstrates a language/runtime boundary that can compile governance metadata, deny unsupported provider execution, and generate internally verifiable local evidence. It does not turn declarations into external truth. Throughout the paper we enforce five separations:

declaration	≠	verification,
digest integrity	≠	policy approval,
local passport metadata	≠	authenticated identity,
provider contract	≠	executable adapter,
EvidenceBundle	≠	external attestation.

These separations are not caveats appended at the end; they define the method.

Related systems occupy adjacent points in the design space. MCP and A2A standardize agent interaction surfaces [1, 10]. NADA and AgentFacts motivate open naming and discovery for agents [15, 16]. DID and VC standards define identifier and credential data models [26, 27]. OPA and Cedar demonstrate policy-as-code approaches [4, 12]; SPIFFE/SPIRE address workload identity [20, 21]; in-toto and SLSA address provenance and supply-chain evidence [19, 23]. `ArgorixLang` does not replace these systems. Its contribution is to compile local governance constructs and evidence boundaries in one agent runtime while marking unresolved assurance properties as unresolved.

The paper asks five explicit research questions:

- RQ1.** How can governance, sovereign identity, and evidence requirements be represented as compiled language constructs?
- RQ2.** How does the `ArgorixLang` pipeline constrain a chatbot request before a provider operation can be planned?
- RQ3.** What evidence does the runtime generate, and how can its internal consistency be checked offline?
- RQ4.** What do the existing runtime sessions demonstrate, and which controlled experiments remain required?
- RQ5.** How can Agent Passports, ATrust, DCP-AI, and future sovereign discovery fit into an open agentic-web architecture?

Our contributions are:

1. **Compiled governance constructs:** a language architecture that represents agents, passports, policies, runtime profiles, provider contracts, and evidence requirements in one validated pipeline.
2. **Fail-closed provider boundary:** a split between declarative provider contracts and executable adapters, with unsupported external-provider execution denied before provider planning.
3. **Typed policy lattice design:** a required v0.2 policy model that separates pass, deny, review, unknown-rule configuration errors, and malformed policy objects instead of collapsing unknown rules into ordinary violations.
4. **Offline evidence boundary:** an EvidenceBundle model that checks bytecode, trace, security-report, and ledger digest consistency while explicitly excluding source integrity, authorship, timestamp, and side-effect absence from the current claim.
5. **Controlled evaluation agenda:** a snapshot analysis plus required controlled-case matrix, ablations, baselines, and claim-boundary taxonomy.

To prevent overreading, we classify every claim into one of four categories from the outset, summarized in table 3 and applied uniformly thereafter. *Implemented* mechanisms include the compiler, the fail-closed VM path, the simulated-only provider boundary, traces, the SecurityReport, the EvidenceBundle, and semantic digest verification. *Declarative* metadata includes passports, `ans_name`, ATrust maps, MCP and A2A bridge declarations, and external-provider contracts. *Proposed* architecture includes Sovereign DNS/ANS resolution, a DID/VC resolver, signed bundles, source-digest binding, and live federation. We explicitly do *not* claim real identity authentication, credential verification, cryptographic handshakes,

blockchain immutability, post-quantum security, legal-compliance certification, universal prompt-injection prevention, or production isolation.

ArgorixLang compilation and evidence architecture

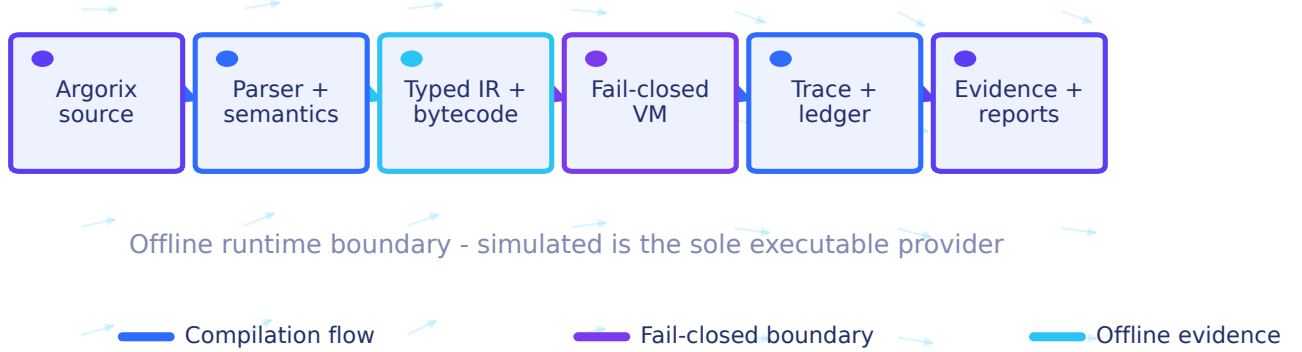


Figure 1: The governed execution and sovereign trust stack. Solid paths denote implemented compilation and runtime behavior; metadata layers do not imply external verification.

Figure 1 summarizes the stack. The empirical result is deliberately narrower than an assurance claim: we observe deterministic structure, bounded execution, and verifiable artifact relationships in one repeated snapshot. We neither conduct a controlled attack experiment nor infer production security from successful digest verification.

Where a figure and table address the same topic, they are deliberately paired rather than duplicative: the figure communicates topology, flow, or boundary placement, while the table provides exact categorical or quantitative comparison. Captions and prose identify the distinct evidentiary role of each.

2 Background and Related Work

2.1 Agent naming and discovery

Project NANDA’s current official title is *Architecting the Internet of AI Agents* [15]. Its research anchor, “Beyond DNS: Unlocking the Internet of AI Agents via the NANDA Index and Verified AgentFacts,” proposes indexing and verifiable facts for networked agents [16]. DNS itself separates hierarchical naming from application semantics [8, 9]. **ArgorixLang** does not implement NANDA, AgentFacts publication, ANS federation, or remote DNS resolution. The comparison is architectural: locally compiled sovereign metadata could become an input to a future discovery service.

2.2 Authenticated delegation and agent authorization

OAuth/OIDC-based agent delegation drafts, FIDO’s Agentic Authentication Working Group, Google AP2, and Mastercard Verifiable Intent all address how agents should request, delegate, authenticate, or evidence user intent in real ecosystems [2, 5, 7, 18]. **ArgorixLang** does not implement these flows. It can declare authorization and credential requirements locally, but it does not authenticate a human, bind a payment mandate, or verify a delegation token.

Table 1: Scope of related-work comparisons. Each row is relational, not an equivalence or conformance claim.

System	Comparison dimension	Bibliographic scope
Project NANDA / AgentFacts / ANS	Agent naming and discovery	Project-level comparison only
ATrust	Agent trust relationships	Project-level comparison only
DCP-AI	Discovery and control plane	Project-level comparison only
OAuth/OIDC, FIDO, AP2, Verifiable Intent	Delegation and agent authorization	Conceptual comparison only
SPIFFE/SPIRE and OWASP NHI	Non-human/workload identity	Scope comparison only
Open Policy Agent	External policy-as-code engine	Direct architectural comparison
Cedar	Authorization policy language	Conceptual comparison only
Jason / AgentSpeak	BDI agent language and interpreter	Direct language/runtime comparison
in-toto / SLSA / VET Your Agent	Evidence, provenance, and external evaluation	Scope comparison only
WASI	Capability-based host isolation	Direct isolation-boundary comparison

2.3 Non-human and workload identity

DID Core and the Verifiable Credentials Data Model define identifier and credential representations [26, 27]. SPIFFE/SPIRE define workload identity mechanisms for software systems [20, 21], and OWASP’s Non-Human Identities Top 10 catalogs operational risks for service accounts, keys, and machine principals [14]. [ArgorixLang](#) passports are local metadata: declaring an identifier or country field is not equivalent to resolving, issuing, or verifying identity.

2.4 Policy-as-code and enforcement

Open Policy Agent separates a general-purpose policy decision engine from application enforcement and expresses decisions in Rego [12]. Cedar offers a policy language oriented around authorization decisions [4]. [ArgorixLang](#) instead compiles policy together with agent, passport, provider, and evidence declarations and emits VM-local decision events. It is not an OPA/Rego or Cedar evaluator.

2.5 Evidence, provenance, and attestation

in-toto records and verifies software supply-chain steps [23]; SLSA defines supply-chain assurance levels and provenance expectations [19]; VET Your Agent studies agent security evaluation as an external assessment problem [6]. [ArgorixLang](#) EvidenceBundles concern one local runtime result, use unsigned semantic digests, and currently exclude source. They do not provide signer authentication, trusted time, TEE attestation, zkTLS/web-proof evidence, or replacement-resistant transparency.

2.6 Agent protocols and capability-based isolation

MCP defines a protocol for context and tool interactions; A2A defines inter-agent communication [1, 10]. WASI specifies standards-track host APIs and a capability-based sandbox model [25]. [ArgorixLang](#) models bridge constraints and records negative execution controls, but it does not claim live MCP/A2A conformance or WASI containment in the current artifact.

Table 1 pairs each system with one comparison dimension. Its rows are side-by-side scope comparisons; neither the table nor the surrounding text claims implementation equivalence, protocol conformance, or shared assurance.

Language evolution and bounded future work

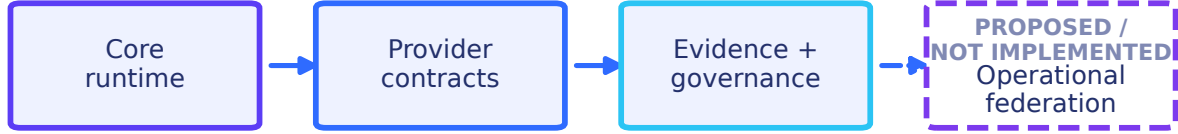


Figure 2: [ArgorixLang](#) language evolution represented by repository versions and feature boundaries. The timeline establishes implementation sequence, not maturity or deployment readiness.

The language is implemented in Rust [17]. The evolution in figure 2 reflects a sequence from agent execution toward typed messages, provider contracts, passports, ATrust metadata, governance, hardening, and evidence. Version numbers establish repository history only; they are not external certifications.

3 Language and Compiler Architecture

[ArgorixLang](#) programs group declarations into modules. Construct families include agents and capabilities, typed messages and handlers, tools and model calls, policies, provider contracts, passports, identity and credential metadata, handshakes, trust ledgers, evidence maps, governance profiles, threat models, runtime profiles, and release/conformance records. Some constructs alter runtime behavior; others are validated descriptions whose execution is explicitly disabled.

Table 4 shows why “represented in the language” is not one claim: typed agents and runtime policy have implemented behavior, whereas ATrust evidence maps and provider contracts remain validated descriptions at the boundary discussed here.

The pipeline has five relevant stages. Parsing produces a source-faithful abstract syntax tree. Name and module resolution bind declarations without discarding provenance. Semantic analysis checks uniqueness, references, cross-construct constraints, and denied combinations. Lowering converts only validated forms to intermediate representation and serializable bytecode. Finally, the VM loads bytecode, installs explicitly executable providers, and evaluates policy before scheduling work.

This order matters. A runtime check alone discovers invalid configuration after deployment; an unchecked lowering pass can erase the distinction between an omitted field and an approved field. Semantic validation rejects, among other cases, unknown references, duplicated names, external execution

without required approval, inconsistent provider/capability bindings, permissive runtime profiles, and trust declarations that assert prohibited security claims. The conformance corpus contains positive and negative examples for these boundaries.

Request lifecycle: controls precede execution

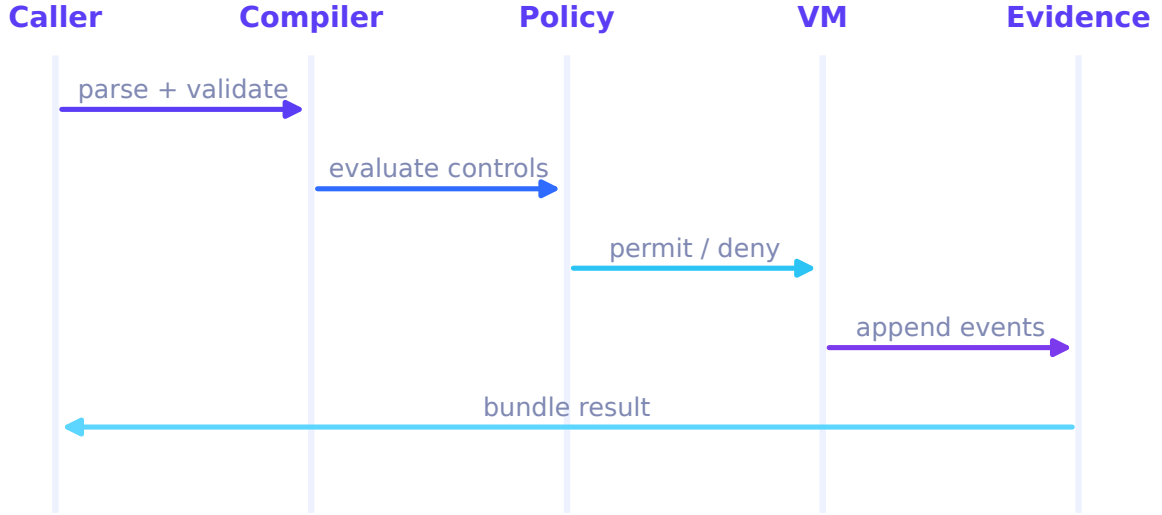


Figure 3: End-to-end request sequence. Semantic validation and policy evaluation precede operation planning; evidence is produced from the resulting runtime state.

Figure 3 answers RQ2 at the architectural level. A request cannot directly select an arbitrary provider. The VM resolves the agent and handler, evaluates policy, checks capabilities, and consults the executable provider registry. Declarative contracts such as `OpenAIPProvider` remain descriptions unless an independently executable adapter is registered. In the observed configuration, the registry contains `simulated`; attempts to cross the external boundary are recorded as blocked.

The language also preserves negative statements. Fields such as `network_allowed=false`, denied key material, mandatory audit, and fail-closed policy are not comments; their combinations are checked. However, the presence of these fields proves only that the compiler accepted the declaration and the runtime recorded its configured handling. It does not prove operating-system containment.

4 Sovereign Identity Metadata

An Agent Passport collects locally asserted identity and governance context: passport name, subject agent, country, jurisdiction, permitted data residency, capabilities, policy bindings, and an `ans_name`. Semantic validation ensures that references resolve and that enumerated relationships are well-formed. Runtime traces can therefore identify which passport and residency constraints accompanied an operation.

The term *sovereign* refers here to explicit jurisdictional and residency metadata controlled by the program, not state recognition or legal status. The observed complete sessions each contain two passports, jurisdiction CL, country CL, and residency values CL and EU. These values are declarations. No external registry validates them.

`ans_name` is implemented as a declarative passport identifier. It survives parsing, semantic validation, lowering, and artifact generation. It is not an operational Agent Naming Service: no DNS query, DID

Declarative trust relationships

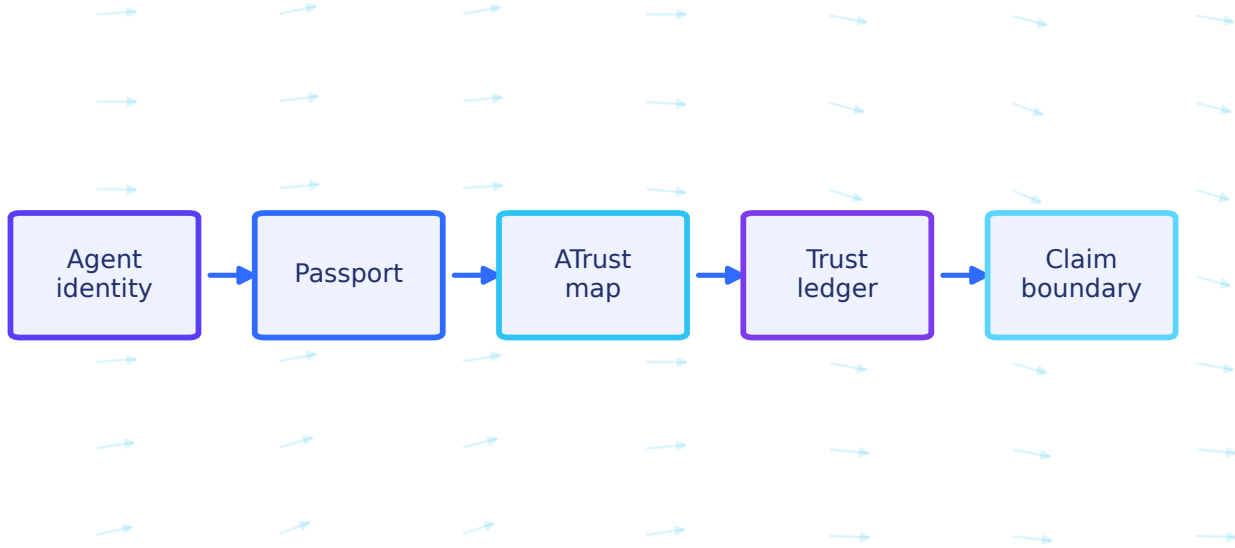


Figure 4: Relationships among passports, agents, policies, ATrust declarations, and evidence. Edges indicate validated references, not verified real-world relationships.

resolution, registry lookup, ownership challenge, or remote authentication occurs. Figure 4 makes the implemented relation explicit: the name is bound to a local passport and agent within the compiled program.

This separation avoids a common category error. A resolvable name answers “where is the record?”; an authenticated identity answers “who controls the relevant key or account?”; a credential answers a scoped assertion; and a policy decision answers whether an operation is permitted in context. A future architecture may connect these functions, but one cannot substitute for another.

Figure 5 places the operational cut immediately after local compilation: federation, DID resolution, and remote proof of control are future components and cannot be inferred from a validated `ans_name`.

5 ATrust and DCP-AI

ATrust is attributed to Edison Vazquez as an unpublished conceptual protocol [24]. In the conceptual model, an agent identity relates to credential descriptions, handshake requirements, trust-ledger entries, and an evidence map. [ArgorixLang](#) implements syntax, semantic relationships, bytecode representations, and trace events for these concepts. It does not implement real credential issuance or verification, cryptographic challenge–response, remote DID resolution, or live attestation.

Identity declarations associate agents with identifier methods and boundaries. Credential contracts describe issuer/subject/type relationships. Handshake declarations describe participants and required evidence. A trust ledger is a local, hash-linked declarative structure whose entries reference subjects and evidence. Evidence maps require named declarations to be covered by artifact links. Validation rejects dangling references and prohibited claims. The word *handshake* therefore denotes a checked contract, not an executed cryptographic exchange; *ledger* denotes a local evidence data structure, not blockchain consensus or immutable storage.

DCP-AI, attributed to Danilo Naranjo Emparanza, proposes a Digital Citizenship Protocol for Au-

Sovereign discovery boundary

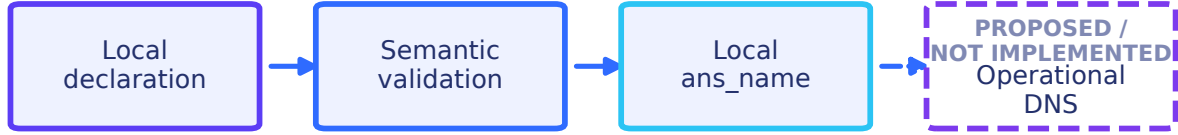


Figure 5: Implemented and proposed discovery boundary. Local `ans_name`, passport, and policy bindings are implemented; Sovereign DNS/ANS federation, DID resolution, and remote discovery are proposed.

onomous AI and a control-plane framing for agents [11]. It complements ATrust’s trust relationships and [ArgorixLang](#)’s compiled enforcement boundary, but the three are not conflated. DCP-AI is a published conceptual proposal; ATrust is unpublished conceptual work; [ArgorixLang](#) is the software artifact evaluated here.

The relationship can be stated operationally. DCP-AI motivates questions of agent standing and governance; ATrust supplies a vocabulary for identity, credentials, handshakes, ledgers, and evidence; [ArgorixLang](#) checks local declarations and emits evidence about their handling. None of these local checks demonstrates that a remote party is genuine.

Figure 6 therefore distinguishes the dashed, unverified source input from the bundle-verified JSON artifacts. Its scope does not extend to an issuer, credential authority, or remote agent.

MCP and A2A bridges follow the same principle. They bind agents, passports, identities, message contracts, and boundaries while requiring network and execution denial in the studied profile. The declarations can support future protocol adapters [1, 10]; today they are declarative boundaries and auditable negative controls.

6 Governed, Fail-Closed Runtime

The runtime evaluates a validated program rather than accepting unrestricted provider calls. Its decision inputs include the requesting agent, capability, target, policy, runtime profile, and executable-provider registry. Denial is the default when a reference is unknown, an approval is absent, a capability is incompatible, or a provider is not executable. Runtime state is retained in an outcome object even when execution fails, allowing a report to describe the denial without converting it into success.

The provider boundary has two registries. Declarative contracts state capabilities, limits, and allowed operations for an external provider. Executable providers contain callable implementations. A contract’s presence does not enroll it for execution. The evaluated chatbot artifacts name an `OpenAIProvider`

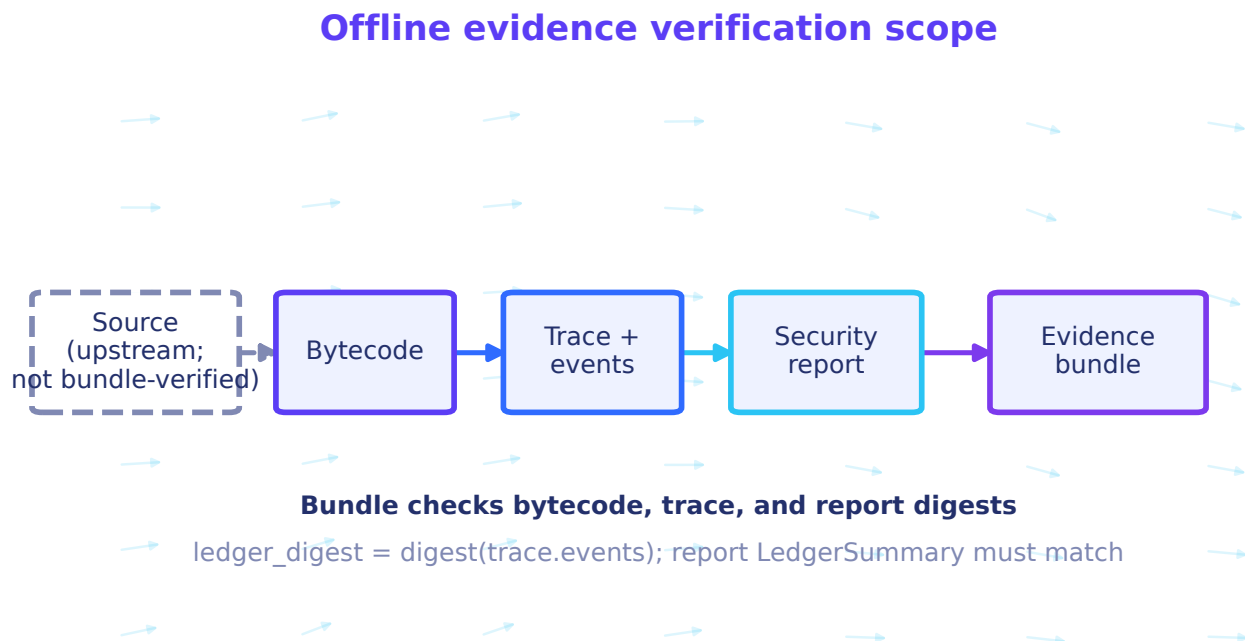


Figure 6: Offline evidence verification scope. Source is upstream and outside bundle verification; verified inputs are bytecode, trace, and security report, with the ledger digest derived from trace events and matched to the report's LedgerSummary.

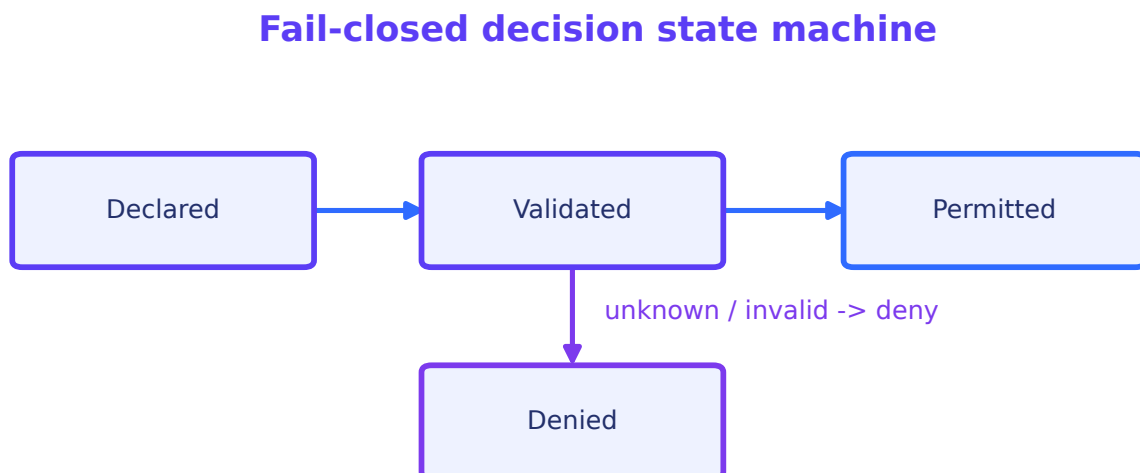


Figure 7: Fail-closed decision state machine. Only validated, policy-permitted operations with an executable provider reach execution; every other branch terminates with an auditable denial or review state.

Table 2: Runtime controls observed in the evaluated profile. These controls describe application-level behavior, not operating-system sandbox proof.

Control	Observed behavior	Scope
Provider execution	simulated only	Implemented
External execution	Blocked	Implemented
Network access	Denied by runtime profiles	Implemented
Secrets and key material	Denied	Implemented

contract and an `OpenAISandbox` descriptor, but the only executable provider is `simulated`. Each complete trace records one blocked external-provider attempt.

Table 2 records the four controls actually observed, while leaving operating-system containment and live-provider behavior outside the table’s scope.

Runtime profiles require audit, evidence, security reports, sandbox metadata, and deny network, plaintext secrets, key material, tools, agent execution, and external execution for the demonstrated path. Governance and threat-model declarations are likewise runtime-disabled. These choices make the corpus suitable for studying compilation and evidence without contacting an external service.

Figure 7 also clarifies the meaning of *fail closed*. It is a control-flow property of the implemented VM: missing authorization does not fall through to a provider call. It is not a claim that every host-language, dependency, operating-system, or deployment path is isolated. A production assessment would need process, filesystem, network, secret-store, and adapter testing beyond the present study.

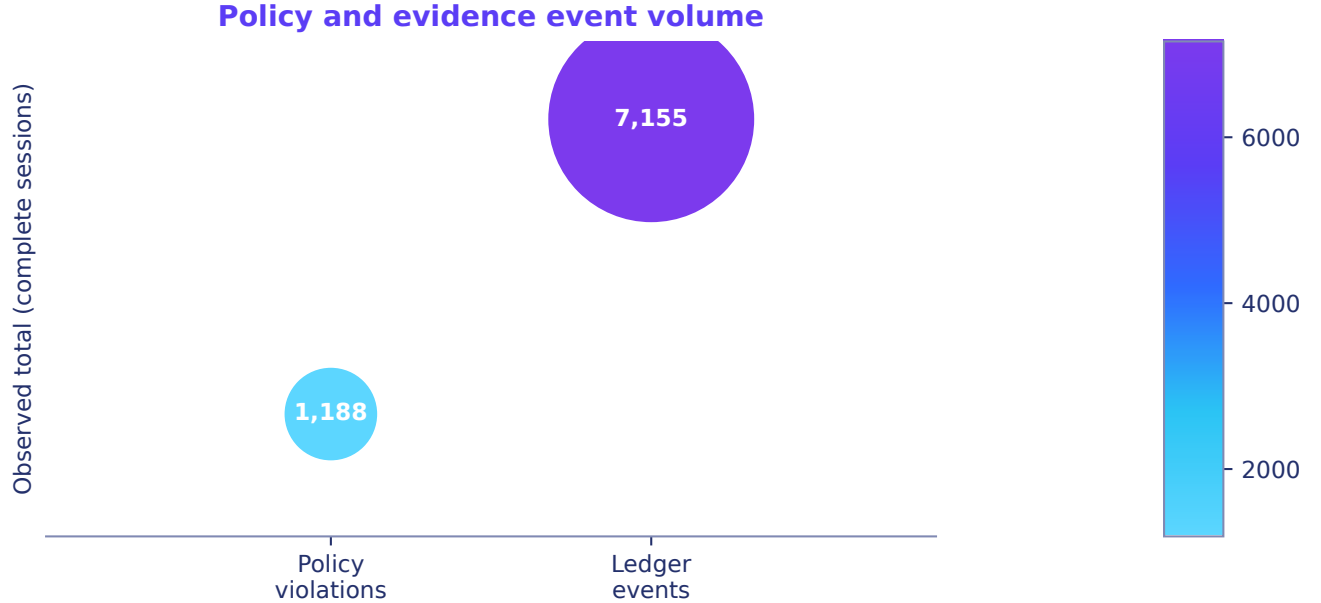


Figure 8: Observed control and policy states across normalized sessions. Blank or unavailable cells for source-only directories are missing observations, not passes.

Figure 8 makes unavailable source-only observations visually distinct from favorable results; this prevents missing artifacts from being counted as control passes. Policies can activate actions, emit violations, and require human review. Coarse status fields are not authoritative when a detailed policy object is available. This rule becomes consequential in section 9: the corpus contains a favorable top-level label alongside uniformly unfavorable detailed policy results.

7 Trace, Report, and Evidence Model

Each complete request directory contains five artifacts: `session.argx`, compiled `session.argbc.json`, `session.trace.json`, `session.security.json`, and `session.evidence.json`. Source is an upstream compiler input, not an EvidenceBundle member. The trace is an ordered ledger of compiler/VM and policy events. The SecurityReport is a structured interpretation of runtime evidence, including execution status, policy results, review state, provider boundary, and declared controls. The EvidenceBundle records four semantic SHA-256 values: `bytecode_digest`, `trace_digest`, `report_digest` for the security report, and `ledger_digest`, which is the canonical digest of `trace.events`. Its artifact map contains `bytecode_path`, `trace_path`, and `security_report_path`; there is no source path or source digest, and there is no standalone ledger path.

Normalized artifact relationships



Figure 9: Schema-level view of the five-artifact request directory. Source is an upstream input outside EvidenceBundle verification; the bundle references bytecode, trace, and security-report semantic digests plus a ledger digest derived from trace events, and does not contain a security verdict.

A semantic digest is computed over canonical JSON derived from a deserialized value, avoiding dependence on whitespace or object-key formatting. The three artifact paths resolve relative to the bundle directory. Offline verification loads bytecode, trace, and the security report and recomputes their three semantic digests. It separately computes `ledger_digest` from `trace.events`. Finally, it checks that the `ledger_digest` stored in the SecurityReport’s LedgerSummary equals the bundle’s `ledger_digest`; the report stores this summary digest, not ledger events. It requires neither network access nor provider credentials. This verifies the relationships encoded by the bundle. Source integrity is outside EvidenceBundle verification.

Figure 9 exposes this asymmetry: a complete request directory contains source, but the bundle verifier starts at compiled bytecode and never asserts that `session.argx` produced it.

Three distinctions are essential. First, a valid digest detects content change relative to the bundle but does not prove who created the bundle. Second, a linked trace does not prove that events outside the instrumented runtime did not occur. Third, artifact integrity does not turn a failing policy result into

Table 3: Interpretive boundaries applied to every result in this study.

Concept	Implemented	Declarative	Proposed	Not claimed
Provider boundary	simulated allowlist	contract metadata	–	external execution
Trust evidence	digest bundle	ATrust map	–	live attestation
Discovery	local ans_name binding/metadata	sovereign meta- data	federation	operational DNS
Security scope	fail-closed controls	threat mappings	deployment study	certification

approval. No signature, trusted timestamp, transparency service, hardware root, or independent witness is evaluated here.

The local trust ledger is similarly bounded. Hash-linked entries and validated references can expose accidental or adversarial modification when the verifier has an expected root or bundle. They do not provide distributed consensus, blockchain immutability, or non-repudiation. The declarations explicitly deny such security claims.

Claim boundary taxonomy

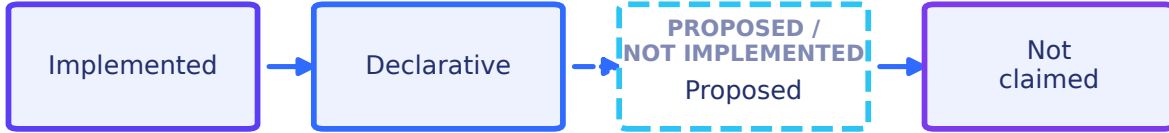


Figure 10: Claim-boundary plate separating implemented mechanisms, declarative metadata, proposed architecture, and properties not claimed.

Figure 10 is the interpretive key for the remainder of the paper: each result is assigned to an implemented, declarative, proposed, or not-claimed region before it is discussed.

Table 3 applies the same distinction to concrete system concepts, preventing digest success from being promoted to live attestation or certification.

Table 4: Representative construct families and their boundary in the studied implementation.

Construct family	Representation	Boundary
Agents and messages	Typed declarations	Implemented
Provider contracts	Declarative metadata	Declarative
Policies and passports	Semantic plus runtime controls	Implemented
ATrust evidence maps	Validated evidence links	Declarative

Table 5: Typed policy decision lattice required for v0.2. The current v0.1 snapshot motivates the UNKNOWN_RULE outcome; it does not yet implement the full lattice.

Outcome	Meaning	Current status
PASS	Known rule satisfied	Required v0.2 outcome
DENY	Known rule violated and execution must stop	Required v0.2 outcome
REVIEW	Known rule requires human review	Required v0.2 outcome
UNKNOWN_RULE	Unsupported policy identifier or configuration error	Diagnosed from v0.1 unknown-rule findings
ERROR	Malformed policy object or parse/semantic failure	Required v0.2 outcome

8 Methodology and System Design

8.1 System boundary

We evaluate [ArgorixLang](#) as a systems artifact for compiled governance, not as a security certification. The implemented path consists of language constructs, semantic validation, typed IR/bytecode, a fail-closed VM, an executable-provider registry, trace generation, a SecurityReport, and an EvidenceBundle verifier. The declarative path consists of Agent Passport fields, ATrust and DCP-AI metadata, external-provider contracts, bridge declarations, and local `ans_name` metadata. Proposed paths include source-digest binding, signed bundles, append-only logs, DID/VC resolution, and Sovereign DNS/ANS federation.

Table 4 is a method table, not a marketing claim: represented constructs are classified by what the current artifact actually does with them.

8.2 Policy decision lattice

The v0.1 snapshot exposes a design defect: all detailed policy findings are reported as “unknown policy rule.” We therefore do not treat those findings as ordinary deny/pass observations. Instead, we specify a v0.2 typed policy decision lattice that future artifacts must implement before policy outcomes can be analyzed as controlled results.

Table 5 separates known pass/deny/review outcomes from configuration errors and malformed policy objects. Required known rules are `provider_executable_allowlist`, `external_provider_without_adapter`, `runtime_network_denied`, `secrets_denied`, `key_material_denied`, `passport_residency_declared`, `passport_residency_conflict`, `source_digest_present`, `bundle_digest_match`, and `evidence_report_consistency`. Until those outcomes are generated, the paper reports policy evidence as a negative diagnostic finding.

Table 6: Committed normalized datasets. Values are regenerated from the authorized artifact snapshot.

Normalized source	Rows / records	Role
runtime_summary.json	33	Session-level normalized summary
sessions.csv	33	Tabular session observations
event_counts.csv	1890	Per-session event-kind counts
verification-results.json	27	Verification result groups

8.3 Snapshot inventory and normalization

The observational input is an authorized snapshot of 33 request directories from the chatbot runtime’s generated output. Publication authorization does not relax minimization: identifiers are aggregated, excerpts are bounded, and prompt content is not reconstructed. A complete directory must have all five expected artifacts listed in section 7; 27 meet that criterion and six contain source only. We retain incomplete directories in the denominator and represent unavailable values as missing.

Table 6 separates session-level normalization, event-level counts, and verifier outcomes so that totals can be recomputed without treating one row type as another. The analysis parses available JSON, records parse errors, counts sizes and artifact presence, extracts execution and policy fields, counts ledger-event kinds, and records passport, runtime-profile, adapter, provider-boundary, and digest metadata. A separate PowerShell procedure invokes the repository’s offline evidence verifier for every complete bundle and stores exit code and bounded standard output.

8.4 Controlled artifact matrix

The snapshot alone is not a behavioral corpus. We therefore define the controlled artifact matrix required to convert the paper from observational evidence into controlled systems evidence. Cases include positive controls, negative controls, review controls, tamper controls, and unknown/error controls. Each case must report expected outcome, observed outcome, policy decision, provider decision, evidence verification, source verification, trace event count, and whether side-effect execution occurred.

Table 7 deliberately distinguishes observed snapshot properties from required experiments. In particular, source-digest matching and source-mismatch detection are v0.2 requirements, because current EvidenceBundles do not verify `session.argx`.

8.5 Ablations and baselines

The ablation plan follows a systems-paper discipline: remove one boundary at a time and measure false allows, false denies, correct denies, correct reviews, tamper detection rate, verification latency, evidence size, trace overhead, compile time, and policy evaluation time. These measurements are not present in the current snapshot and must not be claimed until generated.

Table 8 makes the strongest future claims falsifiable: if removing a boundary does not change a target metric, the paper should not treat that boundary as empirically justified.

Table 9 keeps comparisons fair: [ArgorixLang](#) is scoped to language, policy, provider boundary, and local evidence linkage, while provenance and authorization systems solve different parts of the assurance stack.

8.6 Analysis rules

We define four interpretation rules in advance. A missing artifact is “unavailable,” never “passed.” A detailed policy result overrides a coarser summary label for policy interpretation. Successful evidence

Table 7: Controlled evaluation matrix. Rows marked “Required experiment” or “To be generated” are study requirements, not claimed results.

Case	Expected outcome	Claim status
valid simulated provider plus known policy	PASS; no external side effect	Required experiment
valid passport with residency declared	PASS metadata validation	Required experiment
valid EvidenceBundle	offline verifier passes	Observed for current bundles
valid source digest match	source verification passes	To be generated in v0.2
external provider without executable adapter	DENY	Observed as blocked boundary
network attempt under denied profile	DENY	Observed as denied event
plaintext secret or key material attempt	DENY	Observed as denied class
residency mismatch or unknown jurisdiction	REVIEW	Required experiment
modified trace / report / byte-code	verifier fails	Required tamper experiment
source mismatch	source verifier fails	To be generated in v0.2
unknown policy rule	UNKNOWN_RULE configuration error	Required v0.2 diagnostic
malformed policy object	ERROR parse/semantic failure	Required experiment

verification means digest and reference consistency only. Declarative records are not counted as externally performed actions.

We aggregate complete-session policy findings and ledger events by summing the normalized per-session fields. We check all four evidence digest values for the `sha256:` prefix and 64 hexadecimal digits. The verification result is successful only when the verifier exit code is zero and its explicit verified flag is true.

8.7 Prompt analysis boundary

Although the user authorized publication, we inspect only the bounded `trace.injected.content` schema field. It is absent from all 27 valid complete traces (0 of 27 present). The six source-only directories contain no trace and are not assessable. Prompt publication is a separate allowlisted operation; its suppression is not evidence of content absence. We therefore perform no prompt-level qualitative analysis, quote no prompts, and report no prompt-injection attack success rate.

9 Results

9.1 Snapshot completeness

Of 33 request directories, 27 (81.8%) contain all expected artifacts and six (18.2%) contain source only. All 27 complete sessions report execution status `completed`; execution status is unavailable for the six incomplete sessions. The distinction matters: the study cannot determine whether those six requests failed during compilation, were interrupted, or were retained before downstream artifacts were written.

Figure 11 keeps the six source-only directories in view, while table 10 reports aggregate totals only

Table 8: Required ablation study. Metrics in the last column are measurement obligations, not current results.

Variant	Removed boundary	Do not claim until measured
Full ArgorixLang v0.2	none	false allow, false deny, review accuracy, tamper rate, latency
w/o typed policy lattice	typed outcomes	unknown-rule collapse and aggregation errors
w/o executable-provider allowlist	adapter gate	external-provider false allows
w/o source digest	source binding	source mismatch detection
w/o ledger digest	trace-event ledger binding	ledger tamper detection
w/o report cross-check	report-to-bundle linkage	summary/detail inconsistency detection
w/o fail-closed default	default denial	missing-authorization false allows
JSON audit logs only baseline	cross-artifact verification	tamper and linkage gaps

Table 9: Baseline comparison by scope. Baselines are not adversaries; they identify which assurance layer each system does or does not cover.

Baseline	What it covers	Gap relative to ArgorixLang
Plain application logs	chronological observations	no compiled policy or digest bundle
JSON audit report	structured summary	no cross-artifact verification
OPA/Rego-style external policy	policy decision engine	no compiled passport/evidence constructs unless integrated
in-toto-style provenance	signed supply-chain steps	different scope; Argorix currently unsigned local runtime evidence
MCP/A2A declaration only	protocol surface description	no compiled fail-closed provider boundary

where normalized values exist; together they prevent complete-case counts from becoming a 33-session success claim.

9.2 Deterministic artifact production, limited behavioral diversity

The 27 complete traces contain 7,155 counted ledger events, exactly 265 per complete session. Thus the repeated snapshot shows deterministic artifact production but limited behavioral diversity. It does not show how the system responds to varied prompts, providers, jurisdictions, policy sets, or attacks. The event total is useful for reproducibility checks and evidence-size accounting; it is not evidence of scale in the empirical-security sense.

Each complete trace includes the following denial events: `ExternalProviderExecutionBlocked`, `RuntimeNetworkDenied`, and `RuntimeSecretsDenied`. It also records declaration events for agents, policies, passports, governance, threat, release, ATrust, MCP, and A2A metadata. These observations support the implemented provider-boundary and runtime-denial claims in table 2; they do not prove operating-system containment.

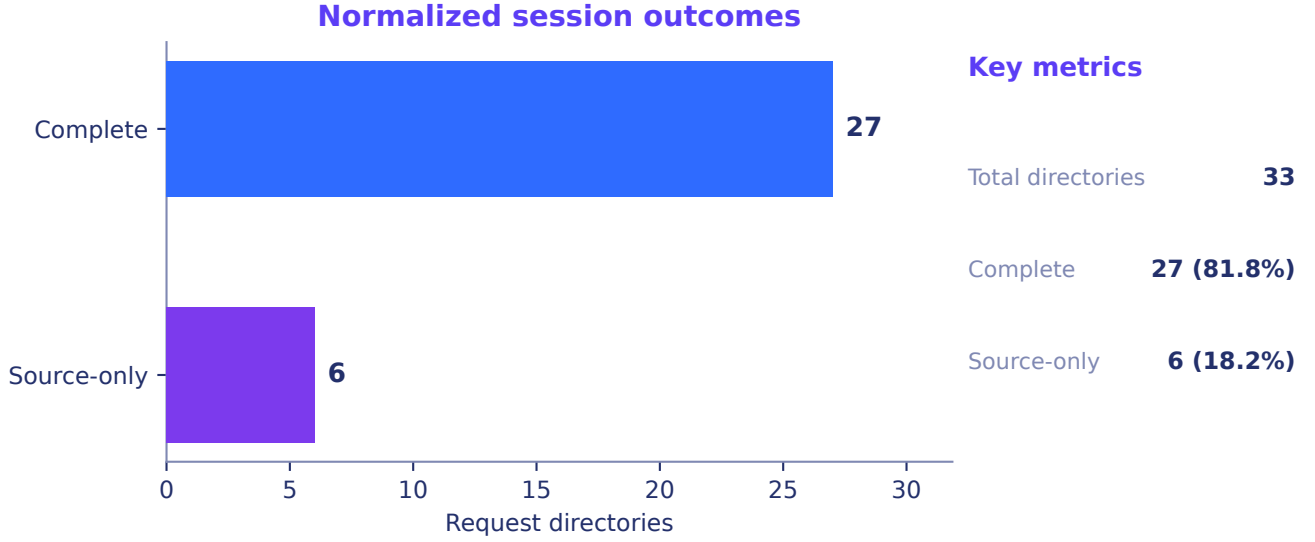


Figure 11: Artifact completeness and observed outcomes for all 33 request directories. Source-only observations remain explicit negative/incomplete results.

Table 10: Empirical totals derived from the normalized snapshot. These totals are descriptive, not broad workload evidence.

Measure	Derived value	Source
Complete sessions	27	runtime_summary.json
Incomplete sessions	6	runtime_summary.json
Normalized sessions	33	sessions.csv
Complete-session policy violations	1188	sessions.csv
Counted ledger events	7155	event_counts.csv
Prompt-content traces inspected / present	27 / 0	runtime_summary.json
Verified evidence bundles	27 / 27	verification-results.json

9.3 Policy diagnostics

Each of the 27 complete sessions has `security_checks=passed` at the top level. Nevertheless, each detailed policy object has `policy_passed=false`, `review_required=true`, and 44 findings whose reason is “unknown policy rule.” The total is therefore 1,188 detailed policy findings.

We treat this as a diagnostic failure of the v0.1 policy-reporting model, not as a normal policy-deny experiment. The correct paper claim is that the current evidence preserves a negative policy state and exposes an aggregation inconsistency. The required v0.2 result is the typed policy decision lattice in table 5, where unknown rules become configuration errors and known rules can produce pass, deny, or review outcomes.

9.4 Offline bundle verification

All 27 complete bundles pass the repository’s offline verifier (27/27). Their bytecode, trace, security-report, and trace-event-ledger digest fields are syntactically valid SHA-256 identifiers. The successful result answers RQ3 narrowly: given each bundle and its relative artifact paths, the verifier can reproduce

Table 11: Principal threats and bounded mitigation claims.

Threat	Mitigation	Claim status
Unauthorized provider execution	Executable-provider allowlist	Implemented
Network side effects	Offline and network-denied pro-files	Implemented
Secret disclosure	Secret and key-material denial	Implemented
Evidence tampering	Digest-linked artifact bundle	Implemented

the semantic digests, recompute the ledger digest from trace events, check the SecurityReport LedgerSummary’s stored digest against it, and validate cross-artifact linkage without network access. The verifier does not digest or verify `session.argx`; Source integrity is outside EvidenceBundle verification.

It does not answer whether policy approved the request; the detailed policy result is negative. It does not authenticate the producer, prove absence of side effects, establish trusted time, or establish semantic correctness of every event. Integrity and approval are orthogonal axes.

9.5 Controlled results still required

Table 7 should be read as the next experimental design, not as completed empirical evidence. The present snapshot supports only a subset of rows: complete bundles verify, external-provider attempts are blocked under the current configuration, network and secret/key-material denial events are present, and source-only directories are visible rather than discarded. The remaining rows—known policy pass/deny/review, source digest match/mismatch, trace/report/bytecode tamper controls, malformed policy errors, and jurisdiction review controls—are required experiments.

9.6 Passports and prompts

Every complete session reports two passports, Chile as country and jurisdiction, and Chilean and European Union residency declarations. These values show that sovereign metadata survives to runtime evidence. They do not show that data was physically stored in either region or that an external authority authenticated the passport.

The bounded `injected.content` field is present in 0 of the 27 complete, valid traces. The six source-only directories have no trace and are not assessable. This presence metric is independent of the publication allowlist: `prompt_text` remains null unless explicitly allowlisted even when content is present. In this snapshot no qualitative prompt themes, quotations, injection labels, or attack success rates can be reported.

10 Security Analysis

We analyze threats as attack paths across trust boundaries, not as proof that a named control defeats every adversary. The relevant assets are authorization state, source and bytecode, policy results, provider selection, credentials and secret material, trace/report integrity, and jurisdictional metadata. Adversaries may supply malformed programs, attempt unauthorized operations, tamper with stored artifacts, exploit a provider adapter, or induce a consumer to accept a favorable summary while ignoring detailed failures.

Figure 12 visualizes which boundaries a threat crosses, and table 11 states the narrower mitigation claim supported at each boundary; neither reports measured adversarial efficacy.

Unauthorized provider execution. The split between declarative and executable registries prevents a provider contract from becoming callable merely by being named. Capability checks, approval require-

Threat-to-control mapping



Figure 12: Threat-mitigation mapping for the evaluated architecture. Mapping means a control is present or declared; it is not a measured attack-blocking rate.

ments, and a simulated-only allowlist reduce the observed attack surface. A malicious executable adapter or VM defect remains in scope for future testing.

Network and secret side effects. The demonstrated runtime profile denies network access, external execution, plaintext secrets, and key material. Events make these decisions visible. Because this study does not instrument the host operating system, the evidence supports VM-level denial behavior, not production-grade isolation.

Artifact tampering. Semantic digests detect changes relative to an EvidenceBundle and avoid formatting-dependent hashes. An attacker able to replace both artifacts and the unsigned bundle can create a self-consistent replacement. Signatures, trusted timestamps, independent publication, and key governance are outside the implementation evaluated here.

Policy confusion. The observed status contradiction is an integrity-of-interpretation threat. A dashboard reading only `security_checks` could present an unsafe conclusion even when the detailed object requires review. Consumers should apply a precedence rule: any explicit failure, unknown rule, or required review prevents an aggregate pass. Future schemas should compute one canonical tri-state decision from typed component results.

Prompt injection and content attacks. Prompt injection is a recognized application risk [13]. [ArgorixLang](#) can constrain resulting operations through capabilities and policy, but this corpus contains no prompt text and no controlled attacks. We make no claim of universal injection prevention and report no success rate.

Identity and trust spoofing. Local semantic checks prevent dangling identity and credential references. They do not establish control of a DID, truth of a credential, or execution of a cryptographic handshake. A future resolver and verifier would introduce new trust anchors, replay concerns, revocation, privacy, and network-failure paths [26, 27].

The strongest supported security statement is therefore conditional: within the observed runtime path, requests are mediated by compiled policy and an executable-provider allowlist, denials are recorded, and complete artifacts are offline-consistent. Security outside that path remains unmeasured.

11 Discussion

RQ1: governance as compilation. The construct inventory shows that identity, residency, provider, policy, governance, threat, and evidence relationships can be represented in one language and rejected before execution when internally inconsistent. The benefit is not that declarations become true; it is that assumptions become typed, referentially checked, and available to both runtime and evidence.

RQ2: constrained request planning. The compiler/VM pipeline and provider split place validation, policy, capability, and registry checks before planning. The observed traces contain blocked external attempts and simulated-only execution. The result supports a fail-closed application architecture, while leaving host isolation untested.

RQ3: evidence and offline consistency. Trace, SecurityReport, and EvidenceBundle serve different functions: event history, interpreted runtime state, and content linkage among bytecode, trace, security report, and the ledger digest derived from trace events. The report’s LedgerSummary stores that digest, not the events. All complete bundles verify offline; the upstream source is not part of that verification. The policy contradiction proves why these artifacts should not be collapsed: valid evidence can faithfully preserve an unfavorable result.

RQ4: demonstrated and undemonstrated behavior. The snapshot demonstrates repeatable artifact production for 27 complete sessions, six visible incomplete sessions, stable control events, and offline digest verification. It does not demonstrate prompt robustness, diverse workloads, live providers, identity authentication, credential verification, cryptographic handshakes, production containment, or compliance.

RQ5: open agentic-web placement. Project NANDA provides a research direction for indexed, verifiable agent facts [15, 16]; DCP-AI frames digital citizenship and governance [11]; ATrust supplies an unpublished conceptual trust vocabulary [24]. [ArgorixLang](#) can compile local metadata that might feed such systems, but it is not their resolver, registry, or trust authority.

This positioning differs from protocol conformance. MCP and A2A define interaction protocols [1, 10]; [ArgorixLang](#) presently checks declarative bridge boundaries with execution disabled. Similarly, DID and VC standards specify interoperable representations [26, 27]; the language stores descriptions without resolving or verifying them.

The policy contradiction is also a design lesson for evidence-consuming systems. “Passed” should not be a free string replicated at multiple levels. A typed lattice such as *deny* > *review* > *allow*, with unknown treated as review or deny, would make aggregation monotone. An EvidenceBundle should then bind both component results and the derived decision, enabling a verifier to recompute the aggregate.

Finally, the repeated 265-event trace suggests that exhaustive metadata can become noisy. Evidence maps and report summaries can improve navigation, but summary must remain traceable to component events. Compact evidence is useful only if it cannot erase negative findings.

Table 12: Limitations and future assurance requirements. These are excluded from current claims unless the required mechanism and evaluation are added.

Limitation	Current status	Future assurance requirement
Live provider execution	Not implemented	sandboxed adapter plus host telemetry
Real identity authentication	Not implemented	DID/VC or equivalent resolver and issuer policy
Source integrity	Outside current bundle verification	source digest and source-to-bytecode binding
Producer authentication	Not implemented	signed bundles and key governance
Replacement detection	Not implemented	append-only log or transparency service
Large workload diversity	Not present	controlled workload matrix and pre-registered metrics

12 Limitations and Claim Boundaries

The dataset is small, single-application, and structurally homogeneous. Its 33 directories are not an independent random sample, and the 27 complete traces have identical event totals. Percentages are descriptive only. Six source-only directories have unknown outcomes, introducing survivor bias if analysis is restricted to complete sessions.

No controlled baseline, adversarial campaign, fault injection, concurrency study, performance benchmark, or usability study was conducted. The bounded prompt-content field was present in 0/27 complete valid traces; six source-only directories were unassessable. Thus prompt-level qualitative analysis and injection measurement are impossible. We do not infer attack success from policy violations: those violations report unknown rules, not successful attacks.

Offline verification checks bytecode, trace, security-report, and trace-event-ledger digest consistency against an unsigned local bundle. It does not check source integrity, authenticate authorship, establish trusted time, prevent replacement of the entire bundle, or prove that the trace is complete. SHA-256 is used for semantic digests; no post-quantum security property is evaluated or claimed.

Passports, `ans_name`, jurisdictions, residency, ATrust identities, credential contracts, handshakes, ledgers, MCP bridges, and A2A bridges are partly or wholly declarative. The current system does not resolve Sovereign DNS/ANS names or DIDs, authenticate real agents, verify credentials, execute cryptographic handshakes, provide blockchain consensus or immutable storage, or run live MCP/A2A interoperability.

Application-level deny flags and the provider registry do not establish production isolation. We did not audit dependencies, prove memory safety of all unsafe/transitive code, inspect operating-system calls, or test container escapes. We make no guarantee against all prompt injection, data leakage, or malicious providers.

Governance mappings are not compliance certification. Neither the language nor this study certifies conformity with a law, regulator, NIST AI RMF, or any other framework [22]. The empirical contradiction between the top-level and detailed statuses further prevents a positive assurance conclusion.

Table 12 converts limitations into concrete future requirements: source integrity needs source-to-bytecode binding, producer authentication needs signatures and key governance, and replacement detection needs an append-only transparency mechanism.

Future evaluations should publish controlled workloads, independent verifiers, OS-level telemetry, and pre-registered outcome definitions.

13 Future Work: Sovereign Discovery

The proposed Sovereign DNS/Agent Naming Service would resolve an `ans_name` to a versioned agent record containing endpoints, passport references, supported protocols, jurisdictional metadata, and evidence locations. It would draw on DNS lessons concerning hierarchical delegation [8, 9], NANDA’s agent-index direction [16], and DID/VC representations [26, 27]. This is proposed architecture, not implemented resolution.

A credible implementation requires more than a lookup API. It needs namespace governance, proof of control, authenticated record updates, revocation, key rotation, replay protection, privacy-preserving queries, caching semantics, conflict handling, auditability, and resilience to compromised registries. Jurisdiction and residency assertions need issuer and evidence policies rather than self-assertion alone.

ATrust contracts could define which identity methods, credential types, and handshake evidence a caller requires. DCP-AI could inform citizenship and control-plane semantics. [ArgorixLang](#) could compile these requirements and reject a discovered record before an adapter is planned. An executable resolver, credential verifier, and cryptographic handshake engine would nevertheless be new trusted components, requiring independent threat models and tests.

Empirically, future work should generate a controlled matrix: valid and invalid records, stale keys, revoked credentials, cross-jurisdiction residency conflicts, unavailable registries, malicious adapters, altered artifacts, and summary/detail contradictions. Outcome measures should include denial correctness, false acceptance, false rejection, verification latency, evidence size, and recovery behavior. Prompt robustness should be studied only with an explicitly collected, privacy-reviewed corpus.

For evidence, signed bundles and append-only transparency publication could provide producer authentication and replacement detection, subject to key governance and availability analysis. This direction should not be described as blockchain immutability unless an actual consensus/storage system is implemented and evaluated.

13.1 Staged roadmap

We sequence the proposed work as a claim-bounded roadmap, in which each stage adds a single trust property and its own evaluation rather than bundling assurance claims. The version labels are planning milestones, not released guarantees.

v0.1 Compiled governance with offline-verifiable EvidenceBundles (the configuration evaluated here).

v0.2 Include a source digest in the EvidenceBundle so that compiled bytecode can be tied back to its source.

v0.3 Signed bundles and an append-only transparency log for producer authentication and replacement detection.

v0.4 A live provider adapter executed under explicit sandbox constraints, with host-level telemetry.

v0.5 A DID/VC resolver and credential verification, introducing the associated trust anchors and revocation paths.

v1.0 Independent, production-oriented evaluation against controlled workloads and pre-registered outcome definitions.

14 Conclusion

[ArgorixLang](#) demonstrates how agent declarations, policy, sovereign metadata, provider boundaries, and evidence requirements can share a compiled systems boundary. In an authorized observational snapshot, 27 of 33 request directories were complete, six were source-only, and all 27 complete EvidenceBundles verified bytecode, trace, security-report, and trace-event-ledger digest consistency offline; source was outside verification. The same complete sessions contained 1,188 detailed policy violations and uniformly

required review despite a favorable top-level status. That contradiction is not incidental: it establishes the paper’s central distinction between integrity and approval.

The implemented contribution is a validated language, fail-closed VM path, simulated-only executable-provider boundary, runtime trace, SecurityReport, EvidenceBundle, and semantic digest verification. Passports and `ans_name` are implemented as declarative identifiers and metadata. ATrust, MCP, and A2A relationships are validated declarations, not live trust or protocol execution. Sovereign DNS/ANS resolution is proposed. Real identity authentication, credential verification, cryptographic handshakes, blockchain immutability, post-quantum security, compliance certification, universal injection prevention, and production isolation are not claimed.

The broader systems lesson is that trustworthy evidence must preserve negative and incomplete outcomes. A verifiable bundle is valuable precisely because it can carry a failed policy decision without relabeling it as success.

Code Availability

The [ArgorixLang](#) compiler, runtime, and offline evidence verifier are openly available. The evaluation in this paper corresponds to a single, archived release.

Repository: <https://github.com/argorixlabs/argorixlang>

Project website: <https://www.argorix-lang.org>

Documentation: <https://www.argorix-lang.org/docs>

Evaluated commit: 6f473f5

Release version: v1.0.1

Code license: Apache License 2.0

Archived DOI: <https://doi.org/10.5281/zenodo.21007563>

The archived DOI resolves to the immutable Zenodo deposit for the release; the repository URL tracks ongoing development and may diverge from the archived state.

Project Website

The project website at <https://www.argorix-lang.org> hosts the language overview, and the documentation portal at <https://www.argorix-lang.org/docs> describes language constructs, the runtime profile, and the evidence schema. Website material is informational and does not extend the claims evaluated here; the archived repository and Zenodo deposit remain authoritative for reproduction.

Artifact Availability

The reproducible build pipeline is contained in the `paper/` directory of the archived repository. It comprises the analyzer that normalizes the runtime snapshot, the PowerShell driver that invokes the offline evidence verifier, the scripts that render every table and vector figure, the static manuscript checks, and the pinned LaTeX build. The committed normalized datasets, generated tables, and generated figures are derived solely from the authorized runtime snapshot and can be regenerated from it. No executable build tool is committed; the LaTeX engine is downloaded on demand and validated against a published checksum before use. Per-request runtime artifacts referenced throughout the paper are the five-artifact directories described in section 7.

Dataset Availability

The empirical study uses an authorized snapshot of 33 runtime request directories generated by the chatbot runtime. The user authorized publication of this material, but authorization does not relax

data minimization. The committed, normalized datasets (session-level, event-level, and verifier-outcome records) contain only aggregated counts and bounded structural fields and may be published in redacted form. Raw prompt text, secrets, key material, provider endpoints, absolute filesystem paths, and other sensitive values are not published. Where the complete corpus is not redistributed, the reason is privacy and data minimization rather than unavailability of results: the normalized records are sufficient to recompute every reported total, while the unredacted inputs remain withheld.

Reproducibility Instructions

The full pipeline normalizes the snapshot, reruns offline bundle verification, regenerates tables and figures, applies static manuscript checks, and renders the preprint. The environment and entry points are as follows; build-host values not fixed by the repository are recorded in the release notes for the archived deposit.

Operating system: developed and validated on Windows; the documented commands target both PowerShell and POSIX shells.

Rust toolchain: as pinned by the repository toolchain configuration.

ArgorixLang version: v1.0.1 (commit 6f473f5).

Python: 3.11 or newer, with the pinned `paper/requirements.txt`.

Input artifact path: `demo/argorix-chatbot-runtime/generated` (relative to the repository root).

Output datasets: `paper/data`, with generated tables and figures in `paper/tables` and `paper/figures`.

The canonical entry point regenerates all artifacts:

```
python -m pip install -r paper/requirements.txt
powershell -NoProfile -ExecutionPolicy Bypass \
  -File paper/scripts/build_paper.ps1 -Target all
```

The offline evidence verifier can be rerun in isolation over the snapshot:

```
powershell -File paper/scripts/verify_runtime.ps1 \
  -InputRoot demo/argorix-chatbot-runtime/generated \
  -OutputPath paper/data/verification-results.json
```

The expected outcome is deterministic: 27 of 33 complete EvidenceBundles verify (27/27 of the complete set), and the six source-only directories are retained as unavailable rather than discarded. Successful verification confirms offline artifact integrity only; it is not policy approval, a security proof, or certification.

Required Artifact Changes Before Expanded Claims

The paper’s stronger v0.2 claims require implementation evidence, not prose alone. Required code and artifact changes are: implement the typed policy decision lattice in table 5; add known rules for executable provider allowlists, external adapters, network denial, secret and key-material denial, passport residency, source digest presence, bundle digest matching, and report consistency; add a source digest to the EvidenceBundle and bind it to the compiled bytecode; generate the controlled cases in table 7; implement tamper tests for modified bytecode, trace, security report, and source; record side-effect execution status per case; and publish latency, evidence-size, trace-overhead, compile-time, and policy-evaluation metrics for the ablations in table 8. Until those artifacts exist, the current paper must report them as Required experiment, To be generated, or Do not claim until measured.

Reviewer Checklist

A reviewer should be able to answer yes to the following bounded claims and no to the corresponding overclaims. The paper claims a compiled governance language/runtime, a fail-closed provider boundary, and offline internal evidence consistency for 27 complete bundles. It does not claim security certification, real identity verification, cryptographic attestation, live ANS/DNS federation, production sandboxing, broad prompt-injection defense, or a large empirical security corpus. Counts over the repeated snapshot are descriptive. Unknown policy rules are treated as a negative diagnostic finding. Source integrity, signatures, append-only logs, workload diversity, and controlled ablations are future assurance requirements unless their artifacts are generated.

Ethics and Privacy Statement

This study analyzes only material that the user authorized for publication, and it applies data minimization throughout. We do not reconstruct prompts, and we publish no secrets, credentials, endpoints, or otherwise sensitive content. The analysis inspects only the bounded `trace.injected.content` schema field, which is present in 0 of 27 complete, valid traces; the six source-only directories contain no trace and are not assessable. Because no prompt content is available, we perform no prompt-level qualitative analysis, quote no prompts, and report no prompt-injection success rate. The absence of prompt text reflects a deliberate minimization decision and the studied runtime configuration; it is not evidence of the absence of risk. The work involves no human subjects, personal data collection, or interaction beyond the authorized artifact snapshot.

Author Contributions

The authors jointly contributed to the conception of the system, the language and runtime design, the implementation under evaluation, the observational analysis, and the writing and revision of the manuscript. Detailed CRediT role assignments will be finalized for the camera-ready version.

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Competing Interests

The authors are affiliated with the organizations developing [ArgorixLang](#) and disclose this relationship as a potential competing interest. The authors declare no other competing financial or non-financial interests.

License

The [ArgorixLang](#) implementation is released under the Apache License 2.0. The archived software deposit is citable through its Zenodo DOI. The license under which this manuscript text is distributed will be confirmed at submission.

A Bounded Artifact Excerpts

The following excerpts are schematic projections of fields present in the normalized data. Values that identify individual requests are omitted; no prompt, secret, endpoint, absolute path, full log, or identifier dump is included.

Listing 1: Redacted session-level projection.

```
{
  "complete": true,
  "artifact_count": 5,
  "execution_status": "completed",
  "security_checks": "passed",
  "policy_passed": false,
  "review_required": true,
  "policy_violation_count": 44,
  "prompt_content_present": false,
  "prompt_text": null
}
```

Listing 2: Redacted provider-boundary projection.

```
{
  "declarative_contract_names": ["OpenAIProvider"],
  "executable_providers": ["simulated"],
  "external_contracts_total": 1,
  "blocked_attempts": 1,
  "external_execution_blocked": true
}
```

Listing 3: Bounded verification-result projection.

```
{
  "exit_code": 0,
  "verified": true,
  "stdout": "Evidence verification: passed",
  "stderr": ""
}
```

B Reproduction Commands

From the root of this paper worktree, the authorized artifact snapshot resides in the primary checkout two levels above. The following commands use only root-relative paths and provide every mandatory analyzer argument:

```
python -m pip install -r paper/requirements.txt
powershell -NoProfile -ExecutionPolicy Bypass -File paper/scripts/build_paper.ps1 -Target all -InputRoot
  ../../demo/argorix-chatbot-runtime/generated
```

In a conventional checkout where generated artifacts are directly under the repository root, replace each `../../demo/` prefix with `demo/`. The runtime analyzer fails visibly if the input is absent. The explicit `-InputRoot` prevents a worktree's tracked but empty `generated/.gitkeep` directory from being mistaken for the authorized snapshot. To rebuild only the PDF with the pinned, checksum-verified Tectonic engine, use:

```
powershell -NoProfile -ExecutionPolicy Bypass -File paper/scripts/build_paper.ps1 -Target paper
```

The committed normalized data permit manuscript and figure regeneration without publishing the original request-directory names or prompt content.

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